

Selenium Revision Notes

Selenium Locators

```
By.id()  
By.name()  
By.linkText()  
By.className()
```

Note:

`By.className()` accepts only a single class name.

✓ Correct

```
By.className("header")
```

✗ Wrong

```
By.className("header section")
```

Use CSS Selector instead:

```
By.cssSelector(".header.section")
```

WebDriver Methods

```
driver.getWindowHandle();    // Returns String  
driver.getWindowHandles();   // Returns Set<String>  
driver.getTitle();  
driver.getCurrentUrl();
```

WebElement Boolean Methods

```
e1.isDisplayed(); // Checks visibility (Logo, Text, Button)
e1.isEnabled();  // Checks whether element is enabled
e1.isSelected(); // Checkbox / Radio button selected
```

Examples:

```
logo.isDisplayed();
username.isEnabled();
radioButton.isSelected();
```

Waits

Implicit Wait

```
driver.manage().timeouts()
    .implicitlyWait(Duration.ofSeconds(10));
```

Explicit Wait (WebDriverWait)

```
WebDriverWait wait =
    new WebDriverWait(driver, Duration.ofSeconds(10));
```

Common Expected Conditions:

```
visibilityOfElementLocated()
elementToBeClickable()
urlContains()
titleContains()
```

Example:

```
wait.until(  
    ExpectedConditions.visibilityOfElementLocated(  
        By.id("username"))));
```

Fluent Wait

Three important things:

1. Timeout Duration
2. Polling Frequency
3. Ignoring Exceptions

```
FluentWait<WebDriver> wait =  
    new FluentWait<>(driver)  
        .withTimeout(Duration.ofSeconds(20))  
        .pollingEvery(Duration.ofSeconds(2))  
        .ignoring(NoSuchElementException.class);
```

Polling Frequency

Polling Frequency = How often Selenium checks for an element or condition before timeout is reached.

Example:

```
.pollingEvery(Duration.ofSeconds(2))
```

Means Selenium checks every 2 seconds.

Navigation Commands

```
driver.navigate().to();  
driver.navigate().forward();  
driver.navigate().back();  
driver.navigate().refresh();
```

Passing URL Object

```
import java.net.URL;

URL url = new URL("https://www.google.com");

driver.navigate().to(url);
```

Window Handling

Single Window ID

```
String parentID = driver.getWindowHandle();
```

Multiple Window IDs

```
Set<String> windowIds =
    driver.getWindowHandles();
```

Convert Set to ArrayList

```
ArrayList<String> list =
    new ArrayList<>(windowIds);
```

Access Window IDs

```
String parentID = list.get(0);
String childID = list.get(1);
```

Switch Window

```
driver.switchTo().window(childID);
```

Close Current Window

```
driver.close();
```

Switch Back to Parent

```
driver.switchTo().window(parentID);
```

Checkboxes

Single Checkbox

```
WebElement singleCheckbox =  
    driver.findElement(By.id("check"));  
  
singleCheckbox.click();
```

Multiple Checkboxes

```
List<WebElement> checkboxes =  
    driver.findElements(By.xpath("//input[@type='checkbox']"));  
  
for(WebElement ch : checkboxes)  
{  
    ch.click();  
}
```

Alerts

```
WebElement e1 =  
    driver.findElement(  
        By.cssSelector("button[id='alertBtn']"));  
  
e1.click();  
  
Alert alertRef =  
    driver.switchTo().alert();
```

Accept Alert

```
alertRef.accept();
```

Dismiss Alert

```
alertRef.dismiss();
```

Send Text

```
alertRef.sendKeys("Hello");
```

Get Alert Text

```
String text = alertRef.getText();
```

Authentication Alerts

Pass username and password directly in URL

```
https://username:password@website.com
```

Example:

```
https://admin:admin123@the-internet.herokuapp.com/basic_auth
```

Frames

Switch to Frame

```
WebElement frame1 =  
    driver.findElement(By.id("frame1"));  
  
driver.switchTo().frame(frame1);
```

Return to Main Page

```
driver.switchTo().defaultContent();
```

Select Class (Dropdown)

```
Select select =  
    new Select(driver.findElement(By.id("country")));
```

Select by Index

```
select.selectByIndex(1);
```

Select by Value

```
select.selectByValue("India");
```

Select by Visible Text

```
select.selectByVisibleText("India");
```

Get All Options

```
List<WebElement> options =  
    select.getOptions();
```

Print All Options

```
for(WebElement option : options)  
{  
    System.out.println(option.getText());  
}
```